Gitte Kremling

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EDUCATION

Ph.D. in Mathematics Aug 2024
University of Wisconsin-Milwaukee, USA
Thesis: "Bootstrap-based goodness-of-fit test for parametric families of conditional distributions"
Supervised by: Gerhard Dikta and Richard Stockbridge
Graduated with highest possible GPA (4.0)

M.Sc. in Applied Mathematics & Computer Science (formerly Technomathematics) Aug 2019
FH Aachen University of Applied Sciences, Germany
Thesis: "Convergence analysis for multi-level spectral deferred corrections"
Graduated with highest possible GPA (1.0)

B.Sc. in Applied Mathematics & Computer Science (formerly Scientific Programming) Aug 2017
FH Aachen University of Applied Sciences, Germany
Graduated with highest possible GPA (1.0)

EMPLOYMENT

Postdoctoral Researcher Apr 2025 – present
University of Hamburg, Germany
Department of Mathematics, PI: Johannes Lederer

Research Associate Sep 2022 – Mar 2025
FH Aachen University of Applied Sciences, Germany
Department of Medical Engineering and Technomathematics

Teaching Assistant Jan 2020 – Jul 2022
University of Wisconsin-Milwaukee, USA
Department of Mathematical Sciences

Research Associate Sep 2017 - Dec 2019
Forschungszentrum Jülich GmbH, Germany
Jülich Supercomputing Center, Division Mathematics and Education

Professional Trainee (Mathematical-Technical Software Developer) Sep 2014 - Aug 2017
Forschungszentrum Jülich GmbH, Germany
Jülich Supercomputing Center, Division Communication Systems

PUBLICATIONS

G. Kremling, J. Näf and J. Lederer (2026). Generative Modeling under Non-Monotonic MAR Missingness via Approximate Wasserstein Gradient Flows. Accepted for publication at NeurIPS 2026. arXiv:2604.04567

G. Kremling and G. Dikta (2026). Asymptotic properties of the MLE in distributional regression under random censoring. *Statistics & Probability Letters, 237*, doi:10.1016/j.spl.2026.110827

G. Kremling and G. Dikta (2025). Bootstrap-based goodness-of-fit test for parametric families of conditional distributions. *Computational Statistics and Data Analysis, 215*, doi:10.1016/j.csda.2025.108289

G. Kremling and R. Speck (2021). Convergence of multilevel spectral deferred corrections. *Communications in Applied Mathematics and Computational Science, 16(2), 227-265*, doi:10.2140/camcos.2021.16.227

PREPRINTS

G. Kremling, F. Iafrate, M. Taheri and J. Lederer (2025, under review). Non-asymptotic error bounds for probability flow ODEs under weak log-concavity. arXiv:2510.17608

SOFTWARE

G. Kremling (2026). FLOWGEM: FLOW-based GEeneration for Missing data [Python code] Available at: https://github.com/gkremling/FLOWGEM.

G. Kremling (2024). gofreg: Bootstrap-Based Goodness-of-Fit Tests for Parametric Regression [R package] Available at: https://CRAN.R-project.org/package=gofreg.

GRANTS, FELLOWSHIPS AND AWARDS

Young Researcher Award Sep 2026
European Conference on Data Analysis
Springer book prize awarded for the best young researcher presentation at the conference

DAAD Travel Grant May 2026
Deutscher Akademischer Austauschdienst, Germany
Granted 990€ for the invited talk at European Meeting of Statisticians, Lugano, Switzerland

SIAM-Funded Selected Participant Jun 2022
Society for Industrial and Applied Mathematics, USA
Granted $700 for travel costs and covered participation for Mathematical Problems in Industry Workshop,Worcester, MA and Graduate Student Mathematical Modeling Camp, Newark, DE, USA

Badge of Honor for B.Sc. and M.Sc. Aug 2017 & Aug 2019
FH Aachen University of Applied Sciences, Germany
Awarded to the top 5% of graduates

Deutschlandstipendium (Fellowship during M.Sc.) Sep 2018 - Aug 2019
FH Aachen University of Applied Sciences, Germany
Provides non-material and financial support (3600€) to high-achieving and committed students

SELECTED PRESENTATIONS

Research Institute for Statistics and Information Science at University of Geneva, Geneva, Switzerland (Invited Seminar Talk) Sep 2026

European Meeting of Statisticians, Lugano, Switzerland (Invited Talk) Aug 2026

Chair of Scientific Computing at Julius-Maximilians-Universität, Würzburg, Germany (Invited Seminar Talk) Jun 2026

Italian Meeting on Probability and Mathematical Statistics, Palermo, Italy (Invited Talk) Jun 2026

Statistical Laboratory, University of Cambridge, Cambridge, UK (Invited Seminar Talk) Apr 2026

Workshop on Stochastic Models, Statistics and Their Applications, Würzburg, Germany (Invited Talk) Mar 2026

International Conference on Statistics and Data Science, Seville, Spain (Invited Talk) Dec 2025

Computational Imaging Group, DESY, Hamburg, Germany (Invited Seminar Talk) Nov 2025

Conference on Mathematics of Machine Learning, Hamburg, Germany (Poster) Sep 2025

Workshop on Mathematical Data Science, Black Forest, Germany (Open question session) Jul 2025

German Probability and Statistics Days, Dresden, Germany (Talk) Mar 2025

Joint Statistical Meetings, Portland, OR, USA (Talk) Aug 2024

European Meeting of Statisticians, Warsaw, Poland (Talk) Jul 2023

TEACHING EXPERIENCE

As Instructor

Statistical Modeling (graduate course), FH Aachen	Fall 2024
Algebra (Upward Bound Math & Science program), UW-Milwaukee	Summer 2022
Precalculus (Upward Bound Math & Science program), UW-Milwaukee	Summer 2022
Calculus and Analytic Geometry I (undergraduate course), UW-Milwaukee	Spring 2022
Algebraic Literacy I and II (undergraduate course), UW-Milwaukee	Fall 2021
Precalculus (undergraduate course), UW-Milwaukee	Spring 2021
Introduction to College Algebra (undergraduate course), UW-Milwaukee	Fall 2020

As Teaching Assistant

Fundamental Concepts of Arithmetic (undergraduate course), Uni Hamburg	Spring 2026
Introduction to Mathematical Thinking and Methods (undergraduate course), Uni Hamburg	Fall 2025
Functional Analysis (graduate course), FH Aachen	Fall 2024
Numerical Methods in Machine Learning (graduate course), FH Aachen	Fall 2024
Bootstrap Methods in Statistics (graduate course), FH Aachen	Spring 2024
Statistical Modeling (graduate course), FH Aachen	Fall 2023
Time Series Analysis and Prognosis (graduate course), FH Aachen	Fall 2023
Numerical Analysis (undergraduate course), FH Aachen	Spring 2019 & 2023
Probability Theory (undergraduate course), FH Aachen	Fall 2019

SERVICE

Member of Faculty Council, Faculty of Mathematics, Informatics and Natural Sciences, University of Hamburg	Oct 2025 – present
Postdoc Network Spokesperson, Hamburg Research Academy	Oct 2025 – Sep 2026

PROFESSIONAL MEMBERSHIPS

American Statistical Association
Bernoulli Society for Mathematical Statistics and Probability
Caucus for Women in Statistics and Data Science
DMV Fachgruppe Stochastik
Institute of Mathematical Statistics
Society for Industrial and Applied Mathematics